VIT

Vellore Institute of Technology

SCHOOL OF ADVANCED SCIENCES

DEPARTMENT OF MATHEMATICS

SL

CONTINUOUS ASSESMENT TEST-I (September 2023)

FALL SEMESTER 2023-24

Programme Name & Branch: B.Tech

Course Code: BMAT205L

Course Name: Discrete Mathematics and Graph theory

Exam Duration: 90 minutes

Maximum Marks:

General instruction(s): Answer all questions 5×10=50

Sl.No.	Question	Mar
1.	Obtain the PCNF and PDNF of $(p \wedge \neg q) \vee (q \wedge \neg p) \vee (r \wedge p)$ by using (i) Truth table technique (ii) Algebraic method	10
2.	Prove that the following premises are inconsistent. $a \to (b \to c)$, $d \to (b \wedge \neg c)$ and $(a \wedge d)$	10
3.	(i) Show that "All flowers are plants. Sunflower is a flower". "Therefore, sunflower is a plant". (ii) Show that $(\forall x)[(B(x) \vee C(x)) \to A(x)] \Rightarrow (\forall x)[B(x) \to A(x)]$ by using the rules of inference for predicate calculus.	5+5
4.	Let $(Z_5, +_5)$ be the group with respect to addition module 5. (i) Form the Cayley's table (ii) find the order of the group (iii) find the identity element of the group (iv) find the order of each element in the group. (v) find the inverse of each element in the group.	10
5.	State and Prove Lagrange theorem on groups.	10